NEET UG HIGH-YIELD BIOLOGY SERIES

Biodiversity and Conservation

Complete NCERT-Aligned Comprehensive Theory, Empirical Equations, Diagrams & Solved Practice Module

Key Learning Objectives for NEET UG

- Differentiate between the three hierarchical levels of biodiversity: genetic, species, and ecological.
- Master global and national biodiversity estimates (IUCN 2004, Robert May's scientific estimate, and India's mega-diversity share).
- Interpret taxonomic distribution among vertebrates, invertebrates, and plants.
- Formulate and evaluate the species-area relationship mathematically and graphically ($S = CA^Z$ and $\log S = \log C + Z \log A$).
- Explain hypotheses governing latitudinal gradients of species richness and the ecological significance of diversity (Tilman's outdoor experiments and Ehrlich's Rivet Popper Hypothesis).
- Examine recent extinctions, vulnerability of taxonomic groups, and the drivers of biodiversity loss (The Evil Quartet).
- Categorize conservation approaches into *in situ* (biodiversity hotspots, protected areas, sacred groves) and *ex situ* strategies (cryopreservation, botanical gardens, tissue culture).

1. Concept and Hierarchical Levels of Biodiversity

Earth displays astonishing biological heterogeneity spanning from intracellular macromolecules to expansive global biomes. The term **biodiversity** was popularised by the renowned sociobiologist **Edward Wilson** to describe the combined diversity existing across all levels of biological organisation.

Level of Diversity	Biological Scope & Defining Attributes	Textbook Examples (Direct NEET Questions)
Genetic Diversity	Variation at the genetic level within a single species across its distributional range. Determines adaptability and phenotypic plasticity.	• *Rauwolfia vomitoria*: Variation in the potency and concentration of the active reserpine chemical across different Himalayan ranges. • India harbours more than **50,000** genetically different strains of rice. • India possesses more than **1,000** varieties of mango.
Species Diversity	The variety and richness of distinct species within a given geographical or ecological region.	The **Western Ghats** have a significantly greater **amphibian species diversity** than the Eastern Ghats.
Ecological (Ecosystem) Diversity	Heterogeneity of habitats, biotic communities, and ecological niches at the biome or ecosystem level.	India (rich in deserts, rain forests, mangroves, coral reefs, wetlands, estuaries, and alpine meadows) displays vastly higher ecosystem diversity than a Scandinavian nation like **Norway**.

2. Global and National Species Wealth

According to the **International Union for Conservation of Nature and Natural Resources (IUCN) (2004)**, the total recorded and described species of plants and animals is slightly more than **1.5 million**. However, global inventory data is profoundly skewed: taxonomic documentation is considerably more thorough in temperate regions than in species-packed tropical realms.

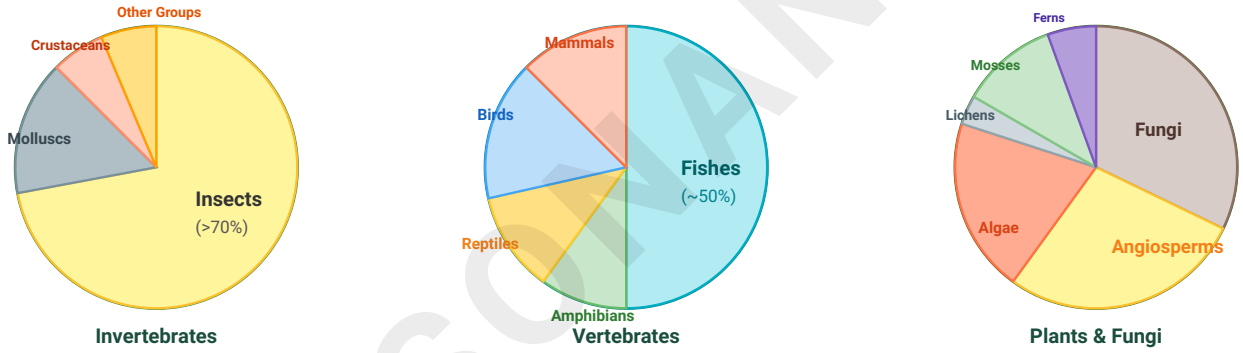
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Extreme estimates of Earth's biological richness range between 20 to 50 million species. A conservative, scientifically sound estimate formulated by **Robert May** places global species diversity at approximately **7 million**.

Proportional Breakdown of Recorded Global Biodiversity

- **Animals:** Comprise more than **70%** of all described species.
- **Plants:** (Including algae, fungi, bryophytes, gymnosperms, and angiosperms) comprise no more than **22%** of the total.
- **Insects:** The most species-rich taxonomic guild among animals, constituting over **70%** of all animal species (out of every 10 animals on Earth, 7 are insects).
- **Fungi:** The number of fungal species worldwide exceeds the combined species sum of fishes, amphibians, reptiles, and mammals.
- **Prokaryotes:** Standard estimates exclude prokaryotic diversity because conventional taxonomic methods are inadequate for microbial characterisation, and many strains cannot be cultured under laboratory conditions. If molecular and biochemical criteria are applied, their diversity alone would extend into millions.

Global Proportionate Species Richness of Major Taxa



Comparative Taxon-Level Biodiversity (Invertebrates, Vertebrates, and Plants/Fungi)

India's Mega-Diversity Status

Although India occupies merely **2.4%** of the world's land surface, its contribution to global species diversity is an impressive **8.1%**, placing it among the **12 mega-diversity nations** of the world.

- Approximately **45,000 species of plants** and nearly **90,000 to 1,00,000 species of animals** have been recorded from India.
- According to Robert May's global estimate, only about **22%** of the total species on Earth have been catalogued thus far.
- Extrapolating this proportion to India indicates that over **1,00,000 plant species** and more than **3,00,000 animal species** remain to be discovered, identified, and described within the country.

3. Patterns of Biodiversity

A. Latitudinal Gradients

Species diversity is not evenly distributed across the globe but exhibits distinct geographic patterns. The most well-documented pattern is the **latitudinal gradient**: species diversity reaches its peak in the tropics and steadily declines toward the poles.

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The tropical belt (situated between $23.5^\circ\,\mathrm{N}$ and $23.5^\circ\,\mathrm{S}$) harbours vastly higher taxonomic richness than temperate or polar zones:

- **Avian Species Richness:** Colombia (equatorial) supports nearly 1,400 bird species; India (predominantly tropical) harbours over 1,200 species; New York ($41^\circ\,\mathrm{N}$) has 105 species; and Greenland ($71^\circ\,\mathrm{N}$) retains only 56 bird species.
- **Vascular Flora:** An equatorial tropical forest in Ecuador contains up to 10 times more vascular plant species than an equal-area forest in the temperate Midwest of the United States.
- **Amazonian Rain Forest (South America):** The richest terrestrial biodiversity repository on Earth. It is home to more than 40,000 species of plants, 3,000 species of fishes, 1,300 species of birds, 427 species of mammals, 427 species of amphibians, 378 species of reptiles, and more than 1,25,000 species of invertebrates. Scientists estimate that at least 2 million insect species remain undescribed in this biome.

Why do the Tropics Harbor Greater Biological Diversity?

1. **Evolutionary Time Hypothesis:** Temperate regions underwent recurrent glaciations throughout geological history, resetting speciation cycles. Tropical latitudes have remained relatively undisturbed for millions of years, providing an uninterrupted, prolonged evolutionary window for speciation and diversification.
2. **Environmental Stability and Constancy:** Unlike temperate environments characterized by severe seasonal shifts, tropical climates are less seasonal, relatively constant, and predictable. Stable conditions promote niche specialization, narrow ecological niches, and higher species co-existence.
3. **Solar Energy and Ecosystem Productivity:** The tropics receive higher incident solar radiation year-round, resulting in elevated primary productivity. Greater available biomass indirectly supports higher diversity and more intricate trophic webs.

B. Species-Area Relationships

During extensive explorations across the wilderness of South American jungles, the German naturalist and geographer **Alexander von Humboldt** observed that within any region, *species richness increases with increasing explored area, but only up to a limit*.

Mathematically, for a wide spectrum of taxa (angiosperms, birds, bats, freshwater fishes), the curve is a **rectangular hyperbola** on normal arithmetic axes:

$$S = CA^Z$$

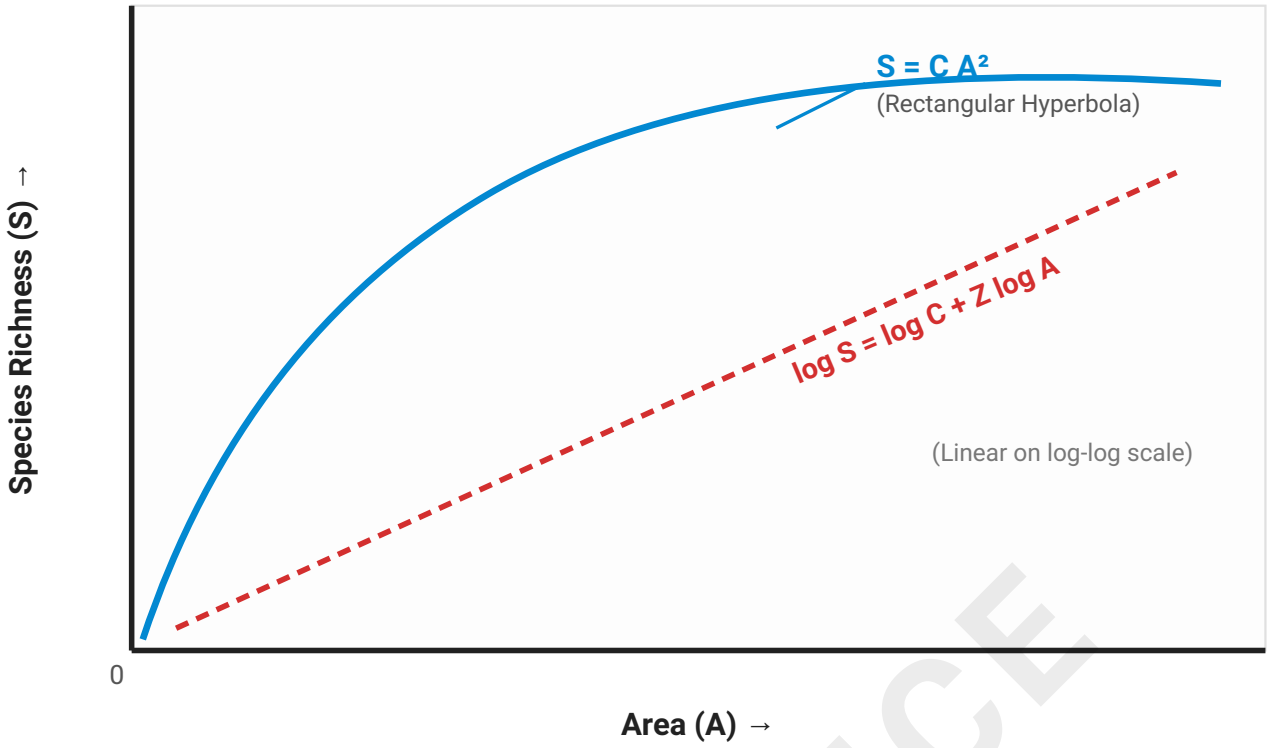
When plotted on a logarithmic (log-log) scale, this relationship is transformed into a linear equation:

$$\log S = \log C + Z \log A$$

Where:

- S = Species richness
- A = Explored Area
- Z = Slope of the line (regression coefficient)
- C = Y-intercept

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Species-Area Relationship on Linear and Logarithmic Scales

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Ecological Significance of the Regression Coefficient (Z)

- **Small to Medium Regional Areas:** Regardless of the taxonomic guild or geographic location (e.g., plants in Britain, birds in California, or molluscs in New York State), the slope Z remains strikingly constant in the range of **0.1 to 0.2**.
- **Entire Continents (Very Large Spatial Scales):** When the species-area relationship is evaluated across continental dimensions, the regression slope becomes much steeper, with Z values ranging between **0.6 and 1.2**.
- **NEET Specific Example:** For frugivorous (fruit-eating) birds and mammals across tropical forests of different continents, the regression slope Z is calculated to be **1.15**.
- *Biological Meaning of a Steeper Slope:* A higher Z value signifies that species richness continues to rise markedly as larger contiguous areas and diverse sub-habitats are sampled.

4. Ecosystem Stability and the Rivet Popper Analogy

Characteristics of a Biologically Stable Community

Long-term ecological research demonstrates that species-rich communities confer higher systemic stability compared to species-poor systems. A stable biological community exhibits three cardinal criteria:

- It must not show extreme year-to-year variation in net primary biomass productivity.
- It must be either resistant or resilient to occasional natural and anthropogenic disturbances.
- It must show resistance to biological invasions by alien/exotic species.

David Tilman's Outdoor Plot Experiments: Tilman evaluated diversity-stability dynamics in long-term field plots and established that:

1. Plots with higher species richness showed less year-to-year variation in total biomass.
2. Increased diversity directly contributed to higher overall productivity.

Paul Ehrlich's 'Rivet Popper Hypothesis'

Stanford ecologist **Paul Ehrlich** presented an intuitive mechanical analogy comparing an ecosystem to an airplane:

Airplane Element (Analogy)	Ecosystem Functional Equivalent
Airplane	The complete ecosystem / biosphere.
Thousands of Rivets	The individual species holding the ecosystem together.
Popping a Rivet	Extinction of a single species caused by human action.
Initial rivet removals	Little immediate impact on flight safety (ecosystem functioning).
Progressive rivet removals	The aircraft becomes dangerously weak over time (ecosystem crash).
Rivets on the wings	**Key/keystone species** that drive fundamental ecosystem dynamics (e.g., primary producers, major predators, key pollinators). Loss of these rivets poses a catastrophic threat.
Rivets on seats or windows	Minor species; loss is comparatively less immediately critical.

5. Biodiversity Loss: Dimensions and Extinctions

Biological wealth is experiencing unprecedented decline directly linked to anthropogenic disturbance:

- Human colonisation of tropical Pacific Islands led to the extinction of over **2,000 species of native birds**.
- The **IUCN Red List (2004)** documents the extinction of **784 species** within the last 500 years, comprising:
  - 338 Vertebrates
  - 359 Invertebrates
  - 87 Plants
- **Notable Recent Extinctions:**
  - **Dodo** (Mauritius)
  - **Quagga** (Africa)
  - **Thylacine** (Australia)
  - **Steller's Sea Cow** (Russia)
  - Three subspecies of tiger: **Bali, Javan, and Caspian**
- Over the past two decades alone, **27 species** have vanished completely.

Vulnerability to Extinction Across Groups

Presently, more than **15,500 species** worldwide are threatened with extinction. Taxonomic vulnerability is structured as follows:

- **Amphibians: 32%** of all species (most vulnerable group).
- **Gymnosperms: 31%** of all species.
- **Mammals: 23%** of all species.
- **Birds: 12%** of all species.

The Sixth Extinction Episode

Earth's fossil record documents **five episodes of mass extinction** since life arose over 3 billion years ago. The ongoing **Sixth Mass Extinction** differs crucially from previous episodes in its **rate**: current extinction rates are estimated to be **100 to 1,000 times faster** than pre-human baseline rates. Ecologists project that at current trajectories, nearly 50% of Earth's species may be extinguished within the next 100 years.

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Consequences of Biodiversity Loss in a Region

1. Decline in overall plant production and primary biomass yield.
2. Lowered resistance to environmental perturbations such as drought.
3. Increased variability in critical ecosystem processes such as water use, plant productivity, and pest and disease cycles.

6. Causes of Biodiversity Loss: 'The Evil Quartet'

Anthropogenic extinction pressures are encapsulated by the classic sobriquet **'The Evil Quartet'**:

1. Habitat Loss and Fragmentation (The Most Dominant Driver)

This represents the single greatest cause driving species to extinction.

- **Tropical Rain Forests:** Once covered over **14%** of the Earth's total land area; they now encompass less than **6%**.
- **The Amazon Rain Forest:** Termed the *'lungs of the planet'*, this biome is cleared rapidly for intensive *soya bean cultivation* and converted to grasslands for beef cattle ranching.
- **Fragmentation:** Large unbroken continuous tracts divided into isolated patches severely disrupt mammals and birds demanding large home ranges and imperil animals exhibiting migratory routines.

2. Over-Exploitation

When human utilization shifts from legitimate survival 'need' to unsustainable economic 'greed', over-exploitation ensues. Historical extinctions driven entirely by over-exploitation include **Steller's sea cow** and the **passenger pigeon**. Currently, many global marine fish stocks are heavily over-harvested.

3. Alien (Exotic) Species Invasions

When non-native species are introduced deliberately or inadvertently, some become aggressively invasive, outcompeting and decimating indigenous biotas:

- **Nile Perch in Lake Victoria (East Africa):** Its introduction caused the catastrophic extinction of an ecologically unique endemic flock of over **200 species of cichlid fish**.
- **Invasive Weed Species:** Carrot grass (*Parthenium*), *Lantana camara*, and water hyacinth (*Eichhornia crassipes*) have aggressively colonised and displaced native flora.
- **African Catfish (*Clarias gariepinus*):** Illegally introduced for aquaculture, it poses an immediate predatory and competitive threat to endemic catfishes in Indian rivers.

4. Co-Extinctions

When an organism becomes extinct, any associate species obligatorily linked to it also faces extinction:

- When a host fish species becomes extinct, its unique obligatory parasite assemblage is automatically extinguished.
- Co-evolved **plant-pollinator mutualisms**: Extinction of a dedicated pollinator insect species inevitably guarantees the extinction of its co-evolved floral partner.

7. Biodiversity Conservation Strategies

Why Should We Conserve Biodiversity?

Category	Core Principle	Illustrative Applications / Data Points
Narrowly Utilitarian	Direct tangible economic benefits extracted by humans from biological resources.	• Direct products: Food (cereals, pulses, fruits), firewood, fiber, construction materials, tannins, resins, dyes, perfumes. • More than **25%** of all prescription pharmaceuticals are derived directly from vascular plants.

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Category	Core Principle	Illustrative Applications / Data Points
		• **25,000 plant species** contribute to indigenous traditional medicines globally. • **Bioprospecting:** Molecular, genetic, and species-level exploration for novel commercially viable biochemical products.
Broadly Utilitarian	Indispensable indirect ecosystem services that sustain biosphere homeostasis.	• The Amazon rain forest alone is estimated to generate **20%** of the Earth's total atmospheric oxygen through photosynthesis. • Pollination of agricultural crops mediated by insect pollinators (bees, bumblebees), birds, and bats. • Climate moderation, soil retention, hydrological buffering, flood mitigation, and intangible aesthetic/spiritual joys.
Ethical	Intrinsic philosophical value possessed by every living species irrespective of utilitarian value.	Every organism has an inherent right to exist; humanity bears a custodial moral duty to protect biodiversity and pass our biological heritage uncompromised to future generations.

How Do We Conserve Biodiversity?

Conservation approaches are systematically divided into two fundamental operational strategies:

1. *In Situ* (On-Site) Conservation

Conservation of species within their natural ecosystems and undisturbed wild habitats ("protecting the entire forest to save the tiger").

- **Biodiversity Hotspots:**
  - Designated regions featuring exceptional **species richness**, high degrees of **endemism** (species restricted solely to that zone and found nowhere else), and severe accelerated habitat destruction.
  - Originally, **25 hotspots** were identified; 9 additional areas were subsequently added, totaling **34 global hotspots**.
  - Together, these 34 hotspots occupy **less than 2%** of Earth's total land area, yet harbour an extraordinary proportion of global biodiversity. Strict legal protection could reduce ongoing mass extinctions by up to **30%**.
  - Three hotspots encompass India's exceptionally biodiverse frontiers:
    1. **Western Ghats and Sri Lanka**
    2. **Indo-Burma**
    3. **Himalaya**

- **Protected Area Network in India:**
  - **14 Biosphere Reserves**
  - **90 National Parks**
  - **448 Wildlife Sanctuaries** (and over 450 as recorded in summary inventories)

- **Sacred Groves:**
  Forest tracts set aside under religious customs where local tribes venerate all vegetation and wildlife, granting them total societal protection:
  - **Khasi and Jaintia Hills** in Meghalaya (vital refuges for rare, threatened botanical taxa)
  - **Aravalli Hills** in Rajasthan
  - **Western Ghat regions** of Karnataka and Maharashtra
  - **Sarguja, Chanda, and Bastar** areas in Madhya Pradesh

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2. *Ex Situ* (Off-Site) Conservation

Threatened, vulnerable, or critically endangered animals and plants are extracted from imperiled natural habitats and placed into specialised facilities under expert human care:

- **Traditional Off-site Enclosures:** Zoological Parks, Botanical Gardens, and Wildlife Safari Parks (vital for maintaining organisms that have become extinct in the wild).
- **Advanced Biotechnological Ex Situ Approaches:**
  - **Cryopreservation:** Storage of viable, fertile gametes in liquid nitrogen ($-196^\circ\,\mathrm{C}$) for prolonged periods.
  - ***In vitro* Fertilization:** Generating embryos under laboratory conditions to propagate endangered animal germplasm.
  - **Plant Tissue Culture:** Micropropagation of imperiled botanical species in sterile synthetic media.
  - **Seed Banks:** Long-term conservation of seeds representing diverse genetic strains of crops and rare wild relatives under controlled low temperature and humidity.

International Treaties and Summits

International Convention	Year & Venue	Core Mandate and Commitments
The Earth Summit (Convention on Biological Diversity - CBD)	**1992** Rio de Janeiro, Brazil	Called upon all signatory nations to implement measures for conservation of biodiversity and ensure the sustainable, equitable utilization of its benefits.
World Summit on Sustainable Development	**2002** Johannesburg, South Africa	**190 nations** pledged their formal commitment to achieve by **2010** a significant reduction in the current rate of biodiversity loss at global, regional, and local levels.

8. Worked Conceptual Examples

Worked Example 1: Calculating Slope and Richness in Species-Area Dynamics

Problem: In a tropical mainland forest, an ecologist models the species-area relationship for frugivorous birds across large continuous tracts. If the baseline equation is $\log S = \log C + Z \log A$, answer the following:

(a) What is the typical empirical value of Z expected for this guild across tropical continental forests?

(b) If the explored continuous forest area is expanded by a factor of 100, by what factor does species richness (S) increase when $Z = 1.15$?

Solution:

(a) According to ecological findings documented in the syllabus, when frugivorous birds and mammals are surveyed across tropical forests of different continents, the regression slope is found to be $Z = 1.15$.

(b) Using the power function:

$$S_1 = CA_1^Z$$

When area increases to $A_2 = 100A_1$:

$$S_2 = C(100A_1)^Z = C(100)^Z A_1^Z = S_1 \times (100)^Z$$

Substituting $Z = 1.15$:

$$\frac{S_2}{S_1} = (10^2)^{1.15} = 10^{2\times 1.15} = 10^{2.30}$$

Since $10^{0.3010} \approx 2$, $10^{2.30} = 10^2 \times 10^{0.30} \approx 100 \times 2 = 200$.

Thus, species richness increases by approximately **200-fold**.

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Worked Example 2: Analyzing Extinction Risk Proportions

Problem: A conservation biology database lists 1,000 representative global vertebrate species comprising 500 fishes, 200 birds, 150 mammals, and 150 amphibians. Based on NCERT global extinction threat statistics, calculate how many birds and amphibians in this cohort face immediate extinction threats.

Solution:
From textbook data on global threat percentages:

- Threatened percentage for birds = **12%**
- Threatened percentage for amphibians = **32%**

Calculations:

- Threatened birds = $200 \times 12\% = 200 \times 0.12 = \mathbf{24}$ species.
- Threatened amphibians = $150 \times 32\% = 150 \times 0.32 = \mathbf{48}$ species.

9. High-Yield NEET UG Practice Questions

Practice Question 1

Single Correct MCQ

The medicinal plant *Rauwolfia vomitoria* growing across different Himalayan ranges displays variability in the potency and concentration of reserpine. This is an example of:

(A) Species diversity

(B) Genetic diversity

(C) Ecological diversity

(D) Biome-level diversity

Answer: (B)
Explanation: Genetic diversity refers to diversity at the genetic level within a single species across its distributional range. Variation in the potency and concentration of reserpine in *Rauwolfia vomitoria*, as well as the 50,000 distinct strains of rice and 1,000 varieties of mango in India, are textbook examples of genetic diversity.

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Practice Question 2

Match The Following

Match the extinct organisms in Column I with their native geographic regions in Column II:

Column I (Organism)	Column II (Native Geographic Region)
a. Dodo	i. Russia
b. Quagga	ii. Australia
c. Thylacine	iii. Mauritius
d. Steller's Sea Cow	iv. Africa

(A) a-iii, b-iv, c-ii, d-i

(B) a-iii, b-i, c-ii, d-iv

(C) a-iv, b-iii, c-i, d-ii

(D) a-ii, b-iv, c-iii, d-i

Answer: (A)
Explanation: Recent extinctions documented by the IUCN Red List include: Dodo (Mauritius), Quagga (Africa), Thylacine (Australia), and Steller's Sea Cow (Russia).

Practice Question 3

Assertion & Reason

Read the assertion and reason statements carefully and select the correct option:

Assertion (A): When species-area relationships are plotted for frugivorous birds across the tropical forests of different continents, the slope of the regression line Z is approximately 1.15.

Reason (R): Over very large spatial scales like entire continents, the regression coefficient Z steepens significantly, lying in the range of 0.6 to 1.2.

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is NOT the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) (A) is false, but (R) is true.

Answer: (A)
Explanation: While the value of Z generally lies between 0.1 and 0.2 when analyzing small or regional areas, surveys over vast continental expanses yield much steeper slopes (Z between 0.6 and 1.2). The value of 1.15 for frugivorous birds and mammals across tropical continental forests is a direct validation of this continental-scale steepening.

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Practice Question 4

Multiple Statement Evaluation

Identify the statement(s) that are **INCORRECT** regarding biodiversity patterns and estimates:

1. Robert May conservatively estimated the total global species diversity to be around 7 million.
2. Fungi have fewer species than the combined total of fishes, amphibians, reptiles, and mammals.
3. More than 70% of all recorded species are animals, and insects constitute over 70% of all animal species.
4. India shares 8.1% of the world's land area and contributes 2.4% to global species diversity.

(A) 1 and 3 only

(B) 2 and 4 only

(C) 2 and 3 only

(D) 1, 2, and 4

Answer: (B)
Explanation: Statement 2 is incorrect because the number of fungal species is *more* than the combined total of fishes, amphibians, reptiles, and mammals. Statement 4 is incorrect because India has **2.4%** of the world's land area and contributes **8.1%** to global species diversity (the numbers are inverted in the statement).

Practice Question 5

Single Correct MCQ

In Paul Ehrlich's 'Rivet Popper Hypothesis', which element of an airplane represents the 'keystone species' driving major ecosystem functions?

(A) Rivets inside the passenger cabin seats

(B) Rivets on the wings of the airplane

(C) Rivets fastening the passenger window frames

(D) The exterior paint on the aircraft fuselage

Answer: (B)
Explanation: In Ehrlich's analogy, loss of rivets on the wings (which represent critical keystone species driving major ecosystem functions) poses a direct, catastrophic threat to flight safety compared to the loss of rivets on seats or windows.

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Practice Question 6

Two-Statement Analysis

Consider the following two statements:

Statement I: Communities with more species generally show less year-to-year variation in total biomass and exhibit higher net primary productivity.

Statement II: The introduction of the African catfish *Clarias gariepinus* for aquaculture led to the extinction of an endemic assemblage of more than 200 species of cichlid fish in Lake Victoria.

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct, but Statement II is incorrect.

(D) Statement I is incorrect, but Statement II is correct.

Answer: (C)
Explanation: Statement I is correct (demonstrated by David Tilman's outdoor experiments). Statement II is incorrect because the introduction of the **Nile perch** led to the extinction of over 200 species of cichlid fish in Lake Victoria. The African catfish *Clarias gariepinus* was illegally introduced into Indian rivers, posing a threat to indigenous catfishes.

Practice Question 7

Ex Situ vs In Situ Classification

Which among the following conservation strategies is **NOT** an *in situ* conservation method?

(A) Biosphere reserve

(B) Sacred grove

(C) Botanical garden

(D) Wildlife sanctuary

Answer: (C)
Explanation: Botanical gardens, zoological parks, and wildlife safari parks are *ex situ* (off-site) conservation methods where organisms are maintained in specialized artificial facilities outside their natural habitats. Biosphere reserves, sacred groves, and wildlife sanctuaries protect populations within their natural ecosystems (*in situ*).

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Practice Question 8

Single Correct MCQ

Which taxonomic group currently faces the highest percentage of extinction risk worldwide according to the IUCN Red List?

(A) Mammals (23%)

(B) Birds (12%)

(C) Amphibians (32%)

(D) Gymnosperms (31%)

Answer: (C)
Explanation: Amphibians are the most vulnerable group, with 32% of all amphibian species currently facing extinction threat worldwide, followed by gymnosperms (31%), mammals (23%), and birds (12%).

Practice Question 9

Geographic Fact-Check

Sacred groves of Meghalaya, which serve as the last refuges for a large number of rare and threatened plants, are located in:

(A) Aravalli Hills

(B) Khasi and Jaintia Hills

(C) Western Ghats

(D) Sarguja and Bastar areas

Answer: (B)
Explanation: Sacred groves in Meghalaya are located in the Khasi and Jaintia Hills. The Aravalli Hills are in Rajasthan; the Western Ghats run through Karnataka and Maharashtra; and the Sarguja, Chanda, and Bastar tracts are in Madhya Pradesh.

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Practice Question 10 Single Correct MCQ

At the historic Earth Summit held in Rio de Janeiro in 1992, the principal pledge undertaken by nations was:

(A) Phase out all hydrochlorofluorocarbons under the Montreal Protocol

(B) Take appropriate measures for the conservation of biodiversity and sustainable utilization of its benefits

(C) Achieve a 50% reduction in greenhouse gas emissions by 2010

(D) Construct ex situ cryopreservation seed banks across all 34 global hotspots

Answer: (B)
Explanation: The Convention on Biological Diversity (The Earth Summit, Rio de Janeiro, 1992) called upon all nations to take appropriate measures for the conservation of biological diversity and sustainable utilisation of its benefits. The commitment to achieve a significant reduction in biodiversity loss rate by 2010 was pledged at the 2002 World Summit in Johannesburg.

10. Common Student Pitfalls & Exam Checklist

Frequent Traps in NEET UG

- **Inverting India's Land Area vs. Diversity Share:** India has **2.4%** of global land area but supports **8.1%** of global species diversity. Questions frequently invert these two numbers.
- **Slope Values (Z) Confusion:** Normal regional surveys yield $Z = 0.1 \text{ to } 0.2$. Continental surveys yield $Z = 0.6 \text{ to } 1.2$. Frugivorous birds in tropical continents have $Z = 1.15$. Never mix these ranges.
- **Fungi vs. Vertebrates:** The species count of fungi is *greater* than the combined total of fishes, amphibians, reptiles, and mammals. Do not assume plants or vertebrates outnumber fungi.
- **Invasive Species Associations:**
  - *Nile perch* → Lake Victoria (East Africa), wiped out 200+ cichlid fish species.
  - *Clarias gariepinus* (African catfish) → Indian rivers, threatened native catfish.
  - *Parthenium* (carrot grass), *Lantana*, *Eichhornia* (water hyacinth) → Invasive weed species.
- **Over-Exploitation Examples:** Steller's sea cow and passenger pigeon became extinct due to human *over-exploitation*, whereas the Dodo vanished due to human habitat encroachment on Mauritius.
- **Conservation Numbers for India:** 14 Biosphere Reserves, 90 National Parks, and 448 Wildlife Sanctuaries (NCERT standard text).

11. Quick Revision Summary

- **Edward Wilson:** Popularized the term 'biodiversity' across genetic, species, and ecological levels.
- **Robert May:** Estimated global species richness at approximately 7 million; ~22% documented so far.
- **Global Pie Share:** Animals > 70%, Plants ≤ 22%, Insects > 70% of all animals. Fungi > all vertebrates combined.
- **Alexander von Humboldt:** Discovered species-area hyperbolic curves: $S = CA^Z$ ($\log S = \log C + Z \log A$).
- **David Tilman:** Higher diversity = lower biomass variation year-to-year + higher total productivity.

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- **Paul Ehrlich:** Rivet Popper Hypothesis (Ecosystem = Airplane; Species = Rivets; Wing Rivets = Key/Keystone species).
- **Extinctions:** 784 species in last 500 years (338 vertebrates, 359 invertebrates, 87 plants). Current rates are 100 to 1,000 times faster than pre-human rates (Sixth Extinction).
- **The Evil Quartet:** Habitat loss and fragmentation (most important), Over-exploitation, Alien species invasions, and Co-extinctions.
- **Hotspots:** Total 34 worldwide (<2% land area, can reduce extinctions by ~30%). Three cover India: Western Ghats & Sri Lanka, Indo-Burma, and Himalaya.
- **Summits:** Earth Summit (1992, Rio de Janeiro); World Summit on Sustainable Development (2002, Johannesburg - 190 countries targeted 2010 reduction).

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